

BREADTH FIRST SEARCH

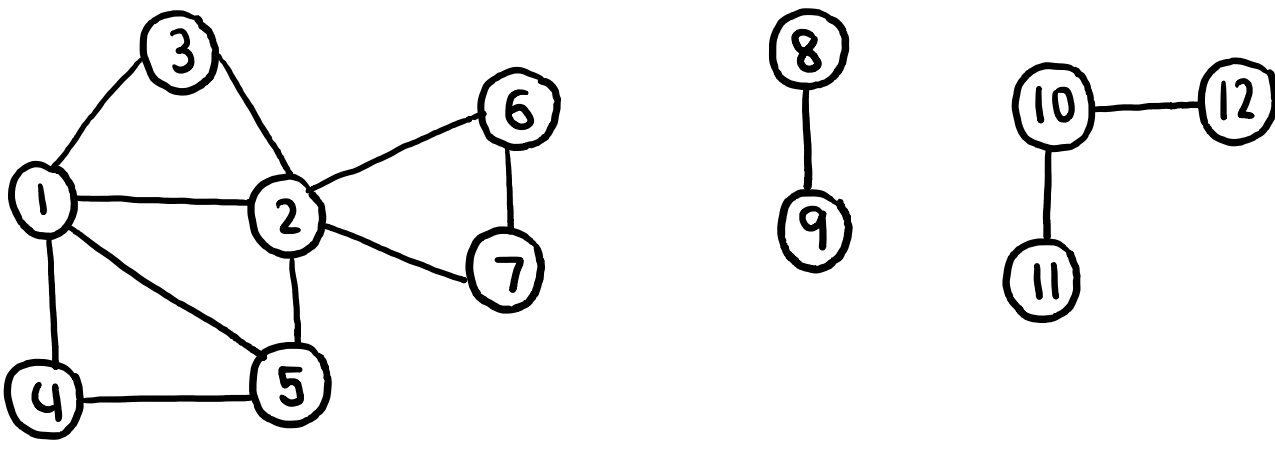
Connected graph: for any pair of vertices $u, v \in V$,
(undirected) there is a path from u to v .
Connectivity can be checked using BFS.

BFS(s):

```
Discovered[s] = True
for all  $v \in V \setminus \{s\}$ :
    Discovered[v] = False
L[0] ← {s}
i ← 0
C ← ∅
While L[i] is not empty:
    L[i+1] ← {}
    for each  $u \in L[i]$ :
        for each edge  $(u,v) \in E$  (incident on  $u$ ):
            if Discovered[v] == False:
                Discovered[v] ← True
                C ← C ∪ {(u,v)}
                L[i+1].append(v)
    i ← i+1
```

For an undirected graph, connectivity = reachability

Q. Compute connected components:



```
Init: for all  $v \in V$ 
    Discovered[v] ← False
Pick a vertex (say  $u$ ).
Run BFS[u] → connected component containing  $u$ .
```

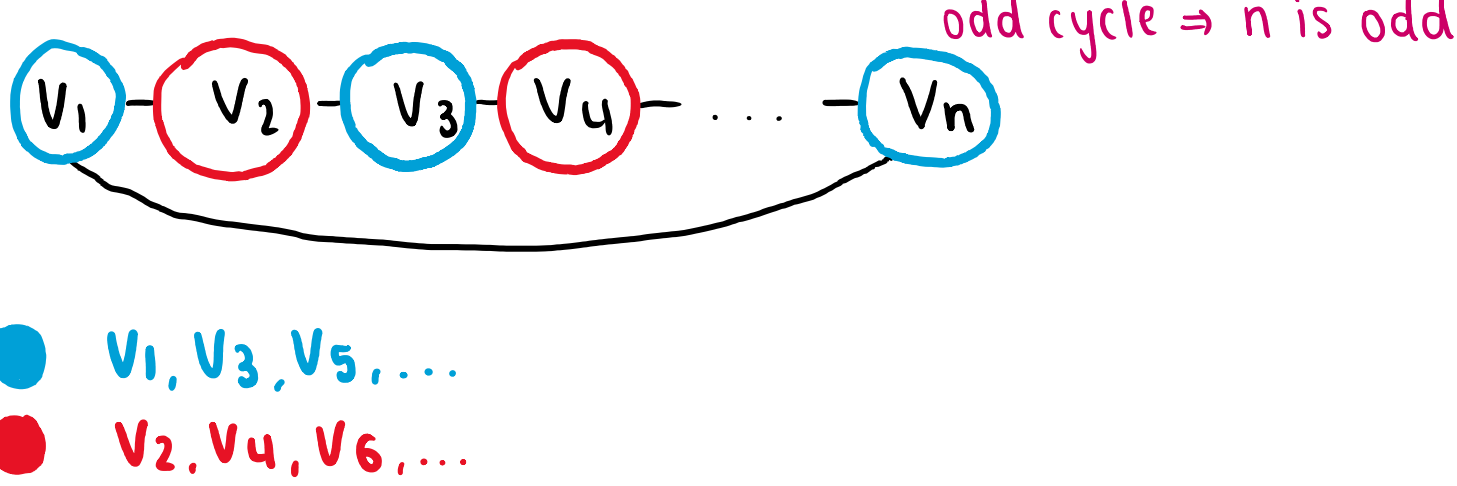
Add bookkeeping for connected components.

• BIPARTITE GRAPH = 2 colourable graph

A graph is bipartite iff it has no odd cycles.
odd no. of edges

Q. Bipartite implies no odd cycles. PROOF:

For the sake of contradiction, assume bipartite graph has an odd cycle.



Since n is odd, V_n is coloured blue.
 $\Rightarrow V_1$ and V_n are both blue.
 $\therefore$ Edge (V_1, V_n) is a monochromatic
contradicting the 2 colorability of the
entire graph.

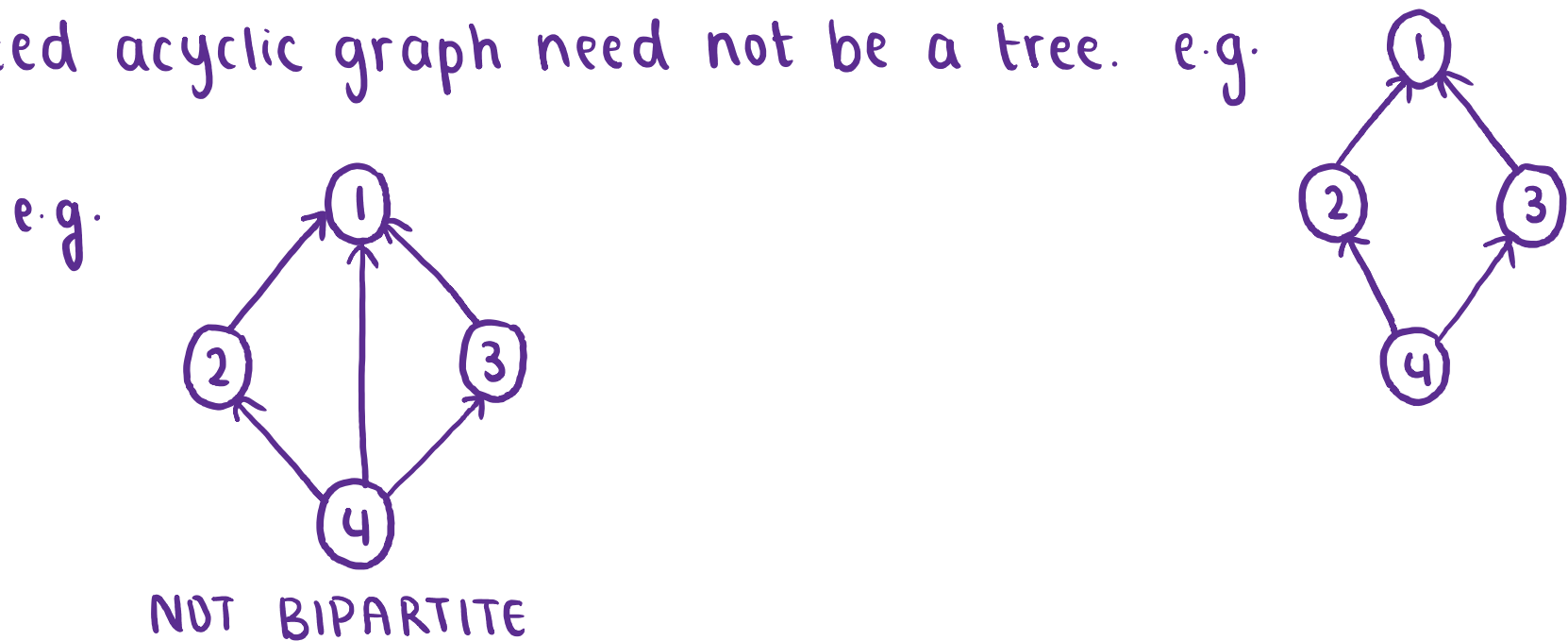
Q. No odd cycles $\Rightarrow$ Bipartite . PROOF:

Start with a vertex.
Every vertex at an odd distance is one colour
and those at even distance are another colour.

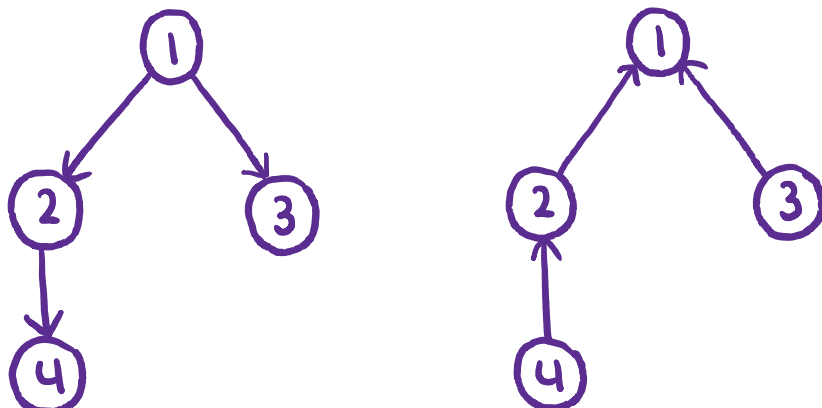
CASE 1: graph is acyclic
 $\Rightarrow$ done
undirected acyclic graph is a tree

CASE 2: there are even cycles
 $\Rightarrow$ continue colouring the cycle w.r.t. the
colouring scheme

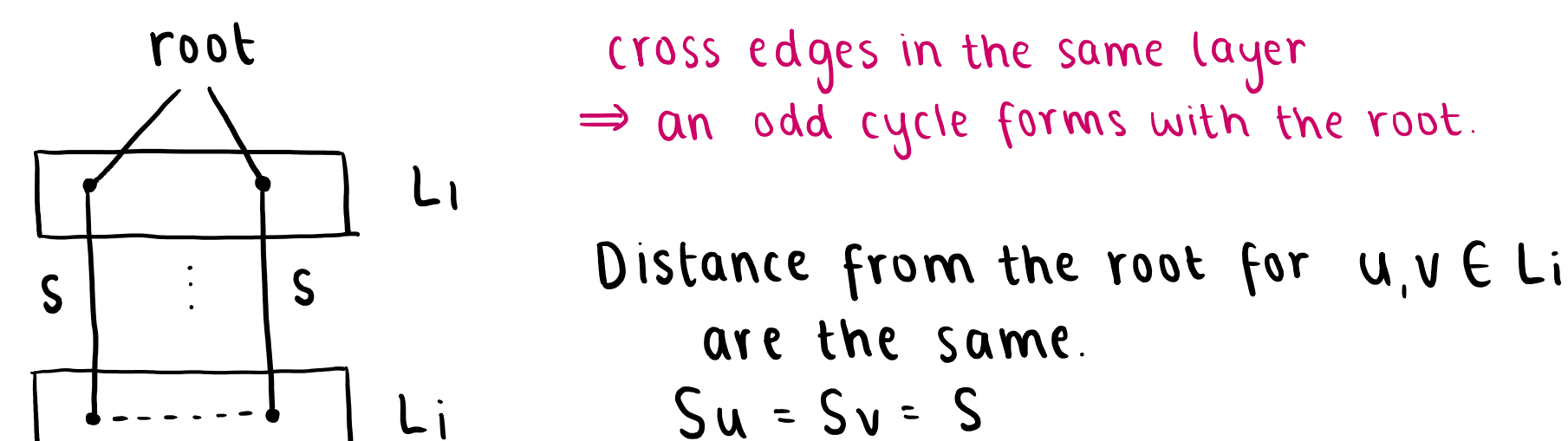
Directed acyclic graph need not be a tree. e.g.



TREE = every node has indegree exactly 1 except root.
(directed)



Q. Suppose we have BFS tree and layers computed
for a given graph. Using this, can we infer bipartiteness?

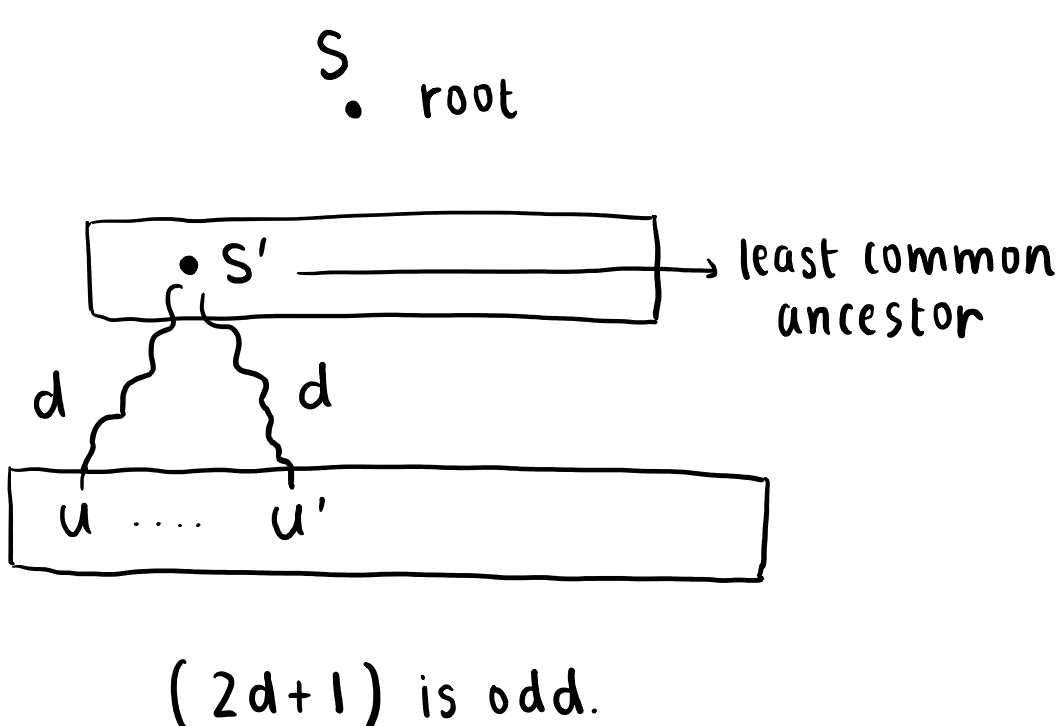


The edge (u,v) contributes to the cycle being an odd cycle.
 $S+1+S = 2S+1$

$\therefore$ Bipartiteness $\rightarrow$ Run BFS
 $\rightarrow$ no cross edges b/w vertices of the same layer.
 $\rightarrow$ assign same colours to alternate layers.

If $(u,v) \quad v \in L_{k-2} \Rightarrow u \notin L_k$
If u and v are neighbors, they must occur either in:
If $(u < v) \Rightarrow u \in L_{k-1}$ or $u \in L_k$
adjacent layer same layer

• Cross edges form an odd cycle with a common ancestor.



BFS
 $\rightarrow$ shortest distance
 $\rightarrow$ connected components
 $\rightarrow$ bipartiteness